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FT - (1)

Module V

Fourier Integral and Fourier Transforms

Fourier series are powerful tools for problems involving periodic functions and are of interest on a finite interval only. But many problems involve functions that are non-periodic and are of interest on the whole x -axis. So extending the method of Fourier series to such functions will lead to Fourier integral.

Fourier Integral representation

If $f(x)$ is defined in $(-\infty, \infty)$, then
$$f(x) = \int_0^{\infty} [A(\omega) \cos \omega x + B(\omega) \sin \omega x] d\omega \quad \text{where}$$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x dx \quad \text{and} \quad B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x dx$$

If x is a point of discontinuity, the above integral is equal to $\frac{1}{2} [f(x^+) + f(x^-)]$

1. Find the Fourier integral representation of the function $f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$

Hence evaluate $\int_0^{\infty} \frac{\sin \omega \cos \omega x}{\omega} d\omega$

Soln : $f(x) = \int_0^{\infty} [A(\omega) \cos \omega x + B(\omega) \sin \omega x] d\omega$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x dx = \frac{1}{\pi} \int_{-1}^1 \cos \omega x dx = \frac{1}{\pi} \left[\frac{\sin \omega x}{\omega} \right]_{-1}^1 = \frac{2 \sin \omega}{\pi \omega}$$

FT - (2)

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x dx = \frac{1}{\pi} \int_{-1}^1 \sin \omega x dx = \frac{1}{\pi} \left[-\frac{\cos \omega x}{\omega} \right]_{-1}^1 = 0$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin \omega}{\omega} \cos \omega x d\omega$$

At $x=1$, the function is ~~dis~~

From the above, $\int_0^{\infty} \frac{\sin \omega \cos \omega x}{\omega} d\omega = \frac{\pi}{2} f(x) = \begin{cases} \frac{\pi}{2} & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$

At $x=1$, the function is discontinuous. Therefore the integral has the value $\frac{f(1^+) + f(1^-)}{2} = \frac{0 + \frac{\pi}{2}}{2} = \frac{\pi}{4}$

2) Using Fourier integral, show that

$$\int_0^{\infty} \frac{\cos \omega x + \omega \sin \omega x}{1 + \omega^2} d\omega = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

Soln: Let $f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x dx = \frac{1}{\pi} \int_0^{\infty} e^{-x} \cos \omega x dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{1 + \omega^2} (-\cos \omega x + \omega \sin \omega x) \right]_0^{\infty}$$

$$= \frac{1}{\pi(1 + \omega^2)}$$

$$\left(\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) \right)$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x dx = \frac{1}{\pi} \int_0^{\infty} e^{-x} \sin \omega x dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{1 + \omega^2} (-\sin \omega x - \omega \cos \omega x) \right]_0^{\infty}$$

$$= \frac{\omega}{\pi(1 + \omega^2)}$$

$$\left(\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) \right)$$

$$\therefore f(x) = \int_0^{\infty} \left[\frac{1}{\pi(1 + \omega^2)} \cos \omega x + \frac{\omega}{\pi(1 + \omega^2)} \sin \omega x \right] d\omega$$

FT-(3)

From the above
$$\int_0^{\infty} \frac{\cos wx + w \sin wx}{1+w^2} dw = \pi f(x) = \begin{cases} \pi e^{-x} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

At $x=0$, the function is discontinuous, therefore the integral has the value $\frac{f(0^+) + f(0^-)}{2} = \frac{\pi + 0}{2} = \underline{\underline{\frac{\pi}{2}}}$

Fourier integral representation of even and odd functions

if $f(x)$ is even in $(-\infty, \infty)$, then $B(w) = 0$

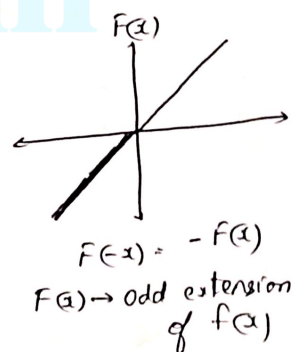
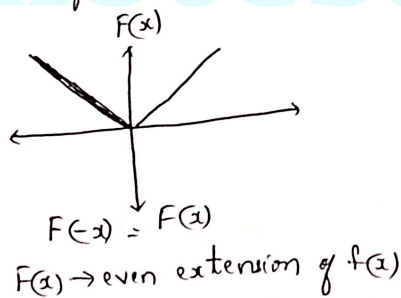
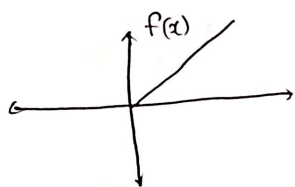
Therefore $f(x) = \int_0^{\infty} A(w) \cos wx dw$ where $A(w) = \frac{2}{\pi} \int_0^{\infty} f(x) \cos wx dx$

if $f(x)$ is odd in $(-\infty, \infty)$, then $A(w) = 0$

Therefore $f(x) = \int_0^{\infty} B(w) \sin wx dw$ where $B(w) = \frac{2}{\pi} \int_0^{\infty} f(x) \sin wx dx$

Fourier Sine Integral and Cosine Integral

if $f(x)$ is a function defined in $(0, \infty)$, it can be extended to an even function or odd function in $(-\infty, \infty)$



Hence in $(0, \infty)$, $f(x)$ can be written as a cosine integral or sine integral.

Cosine Integral

$$f(x) = \int_0^{\infty} A(w) \cos wx dw \quad \text{where} \quad A(w) = \frac{2}{\pi} \int_0^{\infty} f(x) \cos wx dx$$

Sine Integral

$$f(x) = \int_0^{\infty} B(w) \sin wx dw \quad \text{where} \quad B(w) =$$

F7-(4)

1. Represent $f(x) = e^{-kx}$ where $x > 0$, $k > 0$ as a Fourier cosine & sine integral.

Soln :- $A(\omega) = \frac{2}{\pi} \int_0^{\infty} f(x) \cos \omega x dx$

$$= \frac{2}{\pi} \int_0^{\infty} e^{-kx} \cos \omega x dx$$

$$= \frac{2}{\pi} \left[\frac{e^{-kx}}{k^2 + \omega^2} (-k \cos \omega x + \omega \sin \omega x) \right]_0^{\infty}$$

$$= \frac{2}{\pi} \frac{k}{k^2 + \omega^2}$$

$$\therefore f(x) = \frac{2k}{\pi} \int_0^{\infty} \frac{\cos \omega x}{k^2 + \omega^2} d\omega$$

Note :- From the above $\int_0^{\infty} \frac{\cos \omega x}{k^2 + \omega^2} d\omega = \frac{\pi}{2k} e^{-kx}$

$$B(\omega) = \frac{2}{\pi} \int_0^{\infty} e^{-kx} \sin \omega x dx$$

$$= \frac{2}{\pi} \left[\frac{e^{-kx}}{k^2 + \omega^2} (-k \sin \omega x - \omega \cos \omega x) \right]_0^{\infty}$$

$$= \frac{2}{\pi} \frac{\omega}{k^2 + \omega^2}$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\omega \sin \omega x}{k^2 + \omega^2} d\omega$$

Note :- From the above $\int_0^{\infty} \frac{\omega \sin \omega x}{k^2 + \omega^2} d\omega = \frac{\pi}{2} e^{-kx}$

2) Represent $f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$ as a Fourier cosine integral.

FT- (5)

Soln :- $A(\omega) = \frac{2}{\pi} \int_0^1 \cos \omega x \, dx = \frac{2}{\pi} \left[\frac{\sin \omega x}{\omega} \right]_0^1 = \frac{2}{\pi} \frac{\sin \omega}{\omega}$

$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin \omega}{\omega} \cos \omega x \, d\omega$

3) Represent $f(x) = \frac{1}{1+x^2}$, $x > 0$ as a fourier cosine integral.

Soln :- $A(\omega) = \frac{2}{\pi} \int_0^{\infty} \frac{1}{1+x^2} \cos \omega x \, dx$

$$= \frac{2}{\pi} \times \frac{\pi}{2} e^{-\omega} \quad \left[\because \int_0^{\infty} \frac{\cos \omega x}{k^2 + x^2} \, dx = \frac{\pi}{2k} e^{-kx} \right]$$

$$= e^{-\omega}$$

$\therefore f(x) = \int_0^{\infty} e^{-\omega} \cos \omega x \, d\omega$

i) Represent $f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$ as fourier cosine integral.

Soln :- $A(\omega) = \frac{2}{\pi} \int_0^{\pi} \sin x \cos \omega x \, dx = \frac{2}{\pi} \int_0^{\pi} \frac{\sin(1+\omega)x + \sin(1-\omega)x}{2} \, dx$

$$= \frac{1}{\pi} \left[-\frac{\cos(1+\omega)x}{1+\omega} - \frac{\cos(1-\omega)x}{1-\omega} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[-\frac{\cos(1+\omega)\pi}{1+\omega} - \frac{\cos(1-\omega)\pi}{1-\omega} + \frac{1}{1+\omega} + \frac{1}{1-\omega} \right]$$

$$= \frac{1}{\pi} \left[\frac{\cos \pi \omega}{1+\omega} + \frac{\cos \pi \omega}{1-\omega} + \frac{2}{1-\omega^2} \right]$$

$$= \frac{1}{\pi} \left[\frac{2 \cos \pi \omega + 2}{1-\omega^2} \right]$$

$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \pi \omega + 1}{1-\omega^2} \cos \omega x \, d\omega$

FT- (6)

5) Show that
$$\int_0^{\infty} \frac{\cos \frac{\pi \omega}{2}}{1-\omega^2} \cos \omega x d\omega = \begin{cases} \frac{\pi}{2} \cos x & \text{if } 0 < |x| < \frac{\pi}{2} \\ 0 & \text{if } |x| \geq \frac{\pi}{2} \end{cases}$$

Soln: Let
$$f(x) = \begin{cases} \cos x & \text{if } 0 < |x| < \frac{\pi}{2} \\ 0 & \text{if } |x| \geq \frac{\pi}{2} \end{cases}$$

$$A(\omega) = \frac{2}{\pi} \int_0^{\pi/2} \cos x \cos \omega x dx$$

$$= \frac{2}{\pi} \int_0^{\pi/2} \frac{\cos (1+\omega)x + \cos (1-\omega)x}{2} dx$$

$$= \frac{1}{\pi} \left[\frac{\sin (1+\omega)x}{1+\omega} + \frac{\sin (1-\omega)x}{1-\omega} \right]_0^{\pi/2}$$

$$= \frac{1}{\pi} \left[\frac{\sin (1+\omega)\frac{\pi}{2}}{1+\omega} + \frac{\sin (1-\omega)\frac{\pi}{2}}{1-\omega} \right]$$

$$= \frac{1}{\pi} \left[\frac{\cos \frac{\omega\pi}{2}}{1+\omega} + \frac{\cos \frac{\omega\pi}{2}}{1-\omega} \right]$$

$$= \frac{2}{\pi} \frac{\cos \frac{\omega\pi}{2}}{1-\omega^2}$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \frac{\omega\pi}{2}}{1-\omega^2} \cos \omega x d\omega$$

$$\Rightarrow \int_0^{\infty} \frac{\cos \frac{\omega\pi}{2}}{1-\omega^2} \cos \omega x d\omega = \frac{\pi}{2} f(x) = \begin{cases} \frac{\pi}{2} \cos x & \text{if } 0 < |x| < \frac{\pi}{2} \\ 0 & \text{if } |x| \geq \frac{\pi}{2} \end{cases}$$

FT- (7)

6) Represent $f(x)$ as a Fourier Sine integral.

$$f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

Soln :- $B(\omega) = \frac{2}{\pi} \int_0^a x \sin \omega x dx$

$$= \frac{2}{\pi} \left[x \left(\frac{-\cos \omega x}{\omega} \right) + \int \frac{\cos \omega x}{\omega} dx \right]_0^a$$

$$= \frac{2}{\pi} \left[\frac{-x \cos \omega x}{\omega} + \frac{\sin \omega x}{\omega^2} \right]_0^a$$

$$= \frac{2}{\pi} \left[\frac{-a \cos \omega a}{\omega} + \frac{\sin \omega a}{\omega^2} \right]$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \left[\frac{-a \cos \omega a}{\omega} + \frac{\sin \omega a}{\omega^2} \right] \sin \omega x d\omega$$

7) Represent $f(x) = e^x$ if $0 < x < 1$ as Fourier sine integral.
 0 if $x > 1$

Soln :- $B(\omega) = \frac{2}{\pi} \int_0^1 e^x \sin \omega x dx = \frac{2}{\pi} \left[\frac{e^x}{1+\omega^2} (\sin \omega x - \omega \cos \omega x) \right]_0^1$

$$= \frac{2}{\pi} \left[\frac{e}{1+\omega^2} (\sin \omega - \omega \cos \omega) + \frac{\omega}{1+\omega^2} \right]$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \left[\frac{e (\sin \omega - \omega \cos \omega) + \omega}{1+\omega^2} \right] \sin \omega x d\omega$$

8) Show that $\int_0^{\infty} \frac{1 - \cos \pi \omega}{\omega} \sin \omega x d\omega = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$

Soln :- $B(\omega) = \frac{\pi}{2}$ let $f(x) = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$

FT- (8)

$$B(\omega) = \frac{2}{\pi} \int_0^{\pi} \frac{\pi}{2} \sin \omega x dx = \left[-\frac{\cos \omega x}{\omega} \right]_0^{\pi} = -\frac{\cos \pi \omega}{\omega} + \frac{1}{\omega}$$

$$\therefore f(x) = \int_0^{\infty} \left(\frac{1 - \cos \pi \omega}{\omega} \right) \sin \omega x d\omega$$

$$\Rightarrow \int_0^{\infty} \left(\frac{1 - \cos \pi \omega}{\omega} \right) \sin \omega x d\omega = \begin{cases} \frac{\pi}{2} & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

9) show that $\int_0^{\infty} \frac{\sin \pi \omega \sin \omega x}{1 - \omega^2} d\omega = \begin{cases} \frac{\pi}{2} \sin x & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$

Soln:- Let $f(x) = \begin{cases} \sin x & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$

$$\begin{aligned} B(\omega) &= \frac{2}{\pi} \int_0^{\pi} \sin x \sin \omega x dx = \frac{2}{\pi} \int_0^{\pi} \frac{\cos (1-\omega)x - \cos (1+\omega)x}{2} dx \\ &= \frac{1}{\pi} \left[\frac{\sin (1-\omega)x}{1-\omega} - \frac{\sin (1+\omega)x}{1+\omega} \right]_0^{\pi} \\ &= \frac{1}{\pi} \left[\frac{\sin (1-\omega)\pi}{1-\omega} - \frac{\sin (1+\omega)\pi}{1+\omega} \right] \\ &= \frac{1}{\pi} \left[\frac{\sin \pi \omega}{1-\omega} + \frac{\sin \pi \omega}{1+\omega} \right] \\ &= \frac{1}{\pi} \left[\frac{2 \sin \pi \omega}{1-\omega^2} \right] \end{aligned}$$

$$\therefore f(x) = \frac{1}{\pi} \int_0^{\infty} \left(\frac{2 \sin \pi \omega}{1-\omega^2} \right) \sin \omega x d\omega$$

$$\Rightarrow \int_0^{\infty} \frac{\sin \pi \omega \sin \omega x}{1-\omega^2} d\omega = \frac{\pi}{2} f(x) = \begin{cases} \frac{\pi}{2} \sin x & , 0 \leq x \leq \pi \\ 0 & , x > \pi \end{cases}$$

FT- (9)

10) Show that
$$\int_0^{\infty} \frac{\sin w - w \cos w}{w^2} \sin wx dw = \begin{cases} \frac{\pi}{2} x, & 0 < x < 1 \\ \frac{\pi}{4}, & x = 1 \\ 0, & x > 1 \end{cases}$$

Soln:- Let $f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$

$$B(w) = \frac{2}{\pi} \int_0^1 x \sin wx dx = \frac{2}{\pi} \left[x \left(-\frac{\cos wx}{w} \right) + \left(\frac{\cos wx}{w} \right) \right]_0^1$$

$$= \frac{2}{\pi} \left[-x \frac{\cos wx}{w} + \frac{\sin wx}{w^2} \right]_0^1$$

$$= \frac{2}{\pi} \left[-\frac{\cos w}{w} + \frac{\sin w}{w^2} \right]$$

$$= \frac{2}{\pi} \left[-\frac{w \cos w + \sin w}{w^2} \right]$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \left(\frac{\sin w - w \cos w}{w^2} \right) \sin wx dw$$

$$\Rightarrow \int_0^{\infty} \left(\frac{\sin w - w \cos w}{w^2} \right) \sin wx dw = \frac{\pi}{2} f(x)$$

$$= \begin{cases} \frac{\pi}{2} x & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

At $x = 1$, the integral must be equal to $\frac{f(1^+) + f(1^-)}{2}$

$$= \frac{0 + \frac{\pi}{2}}{2} = \underline{\underline{\frac{\pi}{4}}}$$

FT- (10)

H.W

I Represent $f(x)$ as a Fourier Cosine Integral

$$1) f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$2) f(x) = \begin{cases} a^2 - x^2 & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$3) f(x) = \begin{cases} e^{-x} & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

II Represent $f(x)$ as a Fourier Sine Integral

$$1) f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$2) f(x) = \begin{cases} \cos x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$3) f(x) = \begin{cases} e^{-x} & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

III

$$1) \text{ Show that } \int_0^{\infty} \frac{\cos wx}{1+w^2} dw = \frac{\pi}{2} e^{-x} \text{ if } x \geq 0$$

$$2) \text{ Show that } \int_0^{\infty} \frac{w^3 \sin wx}{w^4 + 4} dw = \frac{\pi}{2} e^{-x} \cos x \text{ if } x > 0$$

FT- (11)

Fourier Transform

Fourier transform of $f(x)$ defined in $(-\infty, \infty)$ is given by

$$F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx = \hat{f}(\omega)$$

The original function $f(x)$ is called the inverse Fourier transform of $\hat{f}(\omega)$ and is given by $f(x) = F^{-1}[\hat{f}(\omega)]$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega$$

1. Find the Fourier transform of

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

Soln : $F[f(x)] = \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-i\omega x}}{-i\omega} \right]_{-1}^1$$

$$= \frac{1}{-i\omega\sqrt{2\pi}} [e^{-i\omega} - e^{i\omega}]$$

$$e^{i\omega} = \cos\omega + i\sin\omega \quad \text{and} \quad e^{-i\omega} = \cos\omega - i\sin\omega$$

$$\cancel{e^{i\omega}} - \cancel{e^{-i\omega}} = e^{i\omega} - e^{-i\omega} = -2i\sin\omega$$

$$\therefore \hat{f}(\omega) = \frac{1}{-i\omega\sqrt{2\pi}} (-2i\sin\omega) = \underline{\underline{\frac{\sqrt{2}}{\pi} \frac{\sin\omega}{\omega}}}$$

FT- (Q)

2. Find the Fourier transform of $f(x) = \begin{cases} e^{-ax} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$

$a > 0$

$$\begin{aligned} \text{Soln: } F(e^{-ax}) &= \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-ax} \cdot e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-(a+i\omega)x}}{-(a+i\omega)} \right]_0^{\infty} = \frac{1}{\sqrt{2\pi} (a+i\omega)} \end{aligned}$$

3. Find the Fourier transform of $f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} \text{Soln: } F[f(x)] &= \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^a x \cdot e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[x \frac{e^{-i\omega x}}{(-i\omega)} + \int \frac{e^{-i\omega x}}{i\omega} dx \right]_0^a \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{x \cdot e^{-i\omega x}}{-i\omega} + \frac{e^{-i\omega x}}{\omega^2} \right]_0^a \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{a e^{-i\omega a}}{-i\omega} + \frac{e^{-i\omega a}}{\omega^2} - \frac{1}{\omega^2} \right] \\ &= \frac{1}{\omega^2 \sqrt{2\pi}} \left[(i\omega a + 1) e^{-i\omega a} - 1 \right] \end{aligned}$$

4) Find the Fourier transform of $f(x) = \begin{cases} |x| & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} \text{Soln: } \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 |x| e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\int_{-1}^0 -x e^{-i\omega x} dx + \int_0^1 x e^{-i\omega x} dx \right] \end{aligned}$$

$$\begin{aligned}
 & \text{FT- (13)} \\
 &= \frac{1}{\sqrt{2\pi}} \left\{ \left[-x \left(\frac{e^{-i\omega x}}{-i\omega} \right) - \frac{e^{-i\omega x}}{\omega^2} \right]_0^{-1} + \left[x \left(\frac{e^{-i\omega x}}{-i\omega} \right) + \frac{e^{-i\omega x}}{\omega^2} \right]_0^1 \right\} \\
 &= \frac{1}{\sqrt{2\pi}} \left[\frac{-1}{\omega^2} + \frac{e^{i\omega}}{i\omega} + \frac{e^{i\omega}}{\omega^2} - \frac{e^{-i\omega}}{i\omega} + \frac{e^{-i\omega}}{\omega^2} - \frac{1}{\omega^2} \right] \\
 &= \frac{1}{\sqrt{2\pi}} \left[\frac{-2}{\omega^2} + \frac{2\cos\omega}{\omega^2} + \frac{2i\sin\omega}{i\omega} \right] \\
 &= \sqrt{\frac{2}{\pi}} \left[\frac{\omega\sin\omega + \cos\omega - 1}{\omega^2} \right]
 \end{aligned}$$

5) Find the Fourier transform of

$$f(x) = 1 - x^2 \quad \text{if } |x| < 1$$

$$0 \quad \text{if } |x| > 1$$

Hence evaluate $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx$

$$\text{Soln: } F[f(x)] = \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[(1-x^2) \left(\frac{e^{-i\omega x}}{-i\omega} \right) - (-2x) \frac{e^{-i\omega x}}{(-i\omega)^2} + (-2) \frac{e^{-i\omega x}}{(-i\omega)^3} \right]_{-1}^1$$

$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{2e^{-i\omega}}{\omega^2} - \frac{2e^{-i\omega}}{i\omega^3} - 2\frac{e^{i\omega}}{\omega^2} + 2\frac{e^{i\omega}}{i\omega^3} \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[-\frac{4\cos\omega}{\omega^2} + \frac{4\sin\omega}{\omega^3} \right]$$

$$= \frac{4}{\omega^3 \sqrt{2\pi}} \left[\sin\omega - \omega\cos\omega \right]$$

FT- (14)

Using inverse Fourier transform,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{4}{\omega^3 \sqrt{2\pi}} [\sin \omega - \omega \cos \omega] e^{i\omega x} d\omega$$

$$= \frac{2}{\pi} \int_{-\infty}^{\infty} \left(\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right) (\cos \omega x + i \sin \omega x) d\omega$$

$$\Rightarrow \int_{-\infty}^{\infty} \left(\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right) (\cos \omega x + i \sin \omega x) d\omega = \frac{\pi}{2} f(x)$$

Using even & odd fn property,

$$2 \int_0^{\infty} \left(\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right) \cos \omega x d\omega = \frac{\pi}{2} f(x)$$

$$\Rightarrow \int_0^{\infty} \left(\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right) \cos \omega x d\omega = \frac{\pi}{4} f(x) = \frac{\pi}{4} (1-x^2)$$

Put $x = \frac{1}{2}$

$$\int_0^{\infty} \frac{\sin \omega - \omega \cos \omega}{\omega^3} \cos \frac{\omega}{2} d\omega = \frac{\pi}{4} \left(1 - \frac{1}{4} \right) = \underline{\underline{\frac{3\pi}{16}}}$$

Changing the variable $\omega = x$

$$\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx = \underline{\underline{\frac{-3\pi}{16}}}$$

H.W Find the Fourier transform

$$1) f(x) = \begin{cases} e^{ix} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$2) f(x) = \begin{cases} x e^{-x} & \text{if } -1 < x < 0 \\ 0 & \text{otherwise} \end{cases}$$

$$3) f(x) = e^{-|x|} \quad -\infty < x < \infty$$

FT- (15)

Fourier Cosine and Sine transform

Fourier Cosine transform of $f(x)$ defined in $(0, \infty)$

$$F_c[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx = \hat{f}_c(w)$$

The original function $f(x)$ is called inverse Fourier cosine transform and is given by $f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_c(w) \cos wx \, dw$

Fourier Sine transform of $f(x)$ defined in $(0, \infty)$

$$F_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx \, dx = \hat{f}_s(w)$$

The original function $f(x)$ is called inverse Fourier sine transform and is given by $f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_s(w) \sin wx \, dw$

1. Find the Fourier cosine and sine transform of $f(x) = \begin{cases} k & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$

Soln : $F_c[f(x)] = \hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^a k \cos wx \, dx$

$$= \sqrt{\frac{2}{\pi}} k \left[\frac{\sin wx}{w} \right]_0^a = \sqrt{\frac{2}{\pi}} k \frac{\sin aw}{w}$$

$$F_s[f(x)] = \hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^a k \sin wx \, dx = \sqrt{\frac{2}{\pi}} k \left[-\frac{\cos wx}{w} \right]_0^a$$

$$= \sqrt{\frac{2}{\pi}} k \left[\frac{1 - \cos aw}{w} \right]$$

FT (16)

2) Find the Fourier cosine transform of

$$f(x) = \begin{cases} x & \text{if } 0 < x < 2 \\ 0 & \text{if } x > 2 \end{cases}$$

Soln :- $F_c[f(x)] = \hat{f}_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^2 x \cos \omega x dx$

$$= \sqrt{\frac{2}{\pi}} \left[x \frac{\sin \omega x}{\omega} + \frac{\cos \omega x}{\omega^2} \right]_0^2$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{2 \sin 2\omega}{\omega} + \frac{\cos 2\omega}{\omega^2} - \frac{1}{\omega^2} \right]$$

3) Find the Fourier sine transform of

$$f(x) = e^{-ax}, \quad a > 0$$

Soln :- $F_s[f(x)] = \hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \sin \omega x dx$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{e^{-ax}}{a^2 + \omega^2} (-a \sin \omega x - \omega \cos \omega x) \right) \Big|_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{\omega}{a^2 + \omega^2} \right)$$

4) Find the Fourier sine transform of $f(x) = e^{-|x|}$

Hence evaluate $\int_0^{\infty} \frac{\omega \sin \omega x}{1 + \omega^2} d\omega$

Soln :- $F_s[f(x)] = \hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \sin \omega x dx$ [in (0, \infty) $e^{-|x|} = e^{-x}$]

$$= \sqrt{\frac{2}{\pi}} \left(\frac{e^{-x}}{1 + \omega^2} [-\sin \omega x - \omega \cos \omega x] \right) \Big|_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{\omega}{1 + \omega^2} \right)$$

Using inverse Fourier sine transform,

$$f(x) = \int_0^{\infty} \frac{2}{\pi} \hat{f}_s(\omega) \sin \omega x d\omega$$

FT- (17)

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \sqrt{\frac{2}{\pi}} \frac{\omega}{1+\omega^2} \sin \omega x \, d\omega$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{\omega \sin \omega x}{1+\omega^2} \, d\omega$$

$$\Rightarrow \int_0^{\infty} \frac{\omega \sin \omega x}{1+\omega^2} \, d\omega = \frac{\pi}{2} f(x) = \frac{\pi}{2} e^{-|x|}$$

H.W

1. Find $F_c(e^{-x})$

2. Find $\hat{f}_s(\omega)$ if $f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ -1 & \text{if } 1 < x < 2 \\ 0 & \text{if } x > 2 \end{cases}$

3. Find $\hat{f}_c(\omega)$ if $f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$

Linearity property

$$F[af(x) + bg(x)] = aF[f(x)] + bF[g(x)]$$

Similarly $F_c[af(x) + bg(x)] = aF_c[f(x)] + bF_c[g(x)]$

$$F_s[af(x) + bg(x)] = aF_s[f(x)] + bF_s[g(x)]$$

where a, b are constants.

Fourier transform of derivatives

$$F[f'(x)] = i\omega F[f(x)]$$

$$F[f''(x)] = (i\omega)^2 F[f(x)]$$

Similarly $F[f^{(n)}(x)] = (i\omega)^n F[f(x)]$

FT- (18)

1. Find the Fourier transform of xe^{-x^2} .

Soln :- $F[xe^{-x^2}] = F\left[-\frac{1}{2}(e^{-x^2})'\right]$

$$= -\frac{1}{2} F[(e^{-x^2})']$$

$$= -\frac{1}{2} i\omega F(e^{-x^2})$$

$$= -\frac{1}{2} i\omega \frac{1}{\sqrt{2}} e^{-\omega^2/4}$$

$$= \frac{-i\omega}{2\sqrt{2}} e^{-\omega^2/4}$$

$\left[\because F(e^{-ax^2}) = \frac{1}{\sqrt{2a}} e^{-\omega^2/4a} \right]$

Fourier Cosine & Sine Transform of derivatives

$$\begin{aligned} F_c[f'(x)] &= \omega F_s[f(x)] - \sqrt{\frac{2}{\pi}} f(0) \\ F_s[f'(x)] &= -\omega F_c[f(x)] \end{aligned}$$

$$F_c[f''(x)] = \omega F_s[f'(x)] - \sqrt{\frac{2}{\pi}} f'(0)$$

$$\Rightarrow F_c[f''(x)] = -\omega^2 F_c[f(x)] - \sqrt{\frac{2}{\pi}} f'(0)$$

Similarly $F_s[f''(x)] = -\omega^2 F_s[f(x)] + \sqrt{\frac{2}{\pi}} \omega f(0)$

1. Find the Fourier cosine transform of e^{-ax} using derivatives.

Soln :- $f(x) = e^{-ax}$ $f'(x) = -ae^{-ax}$ $f''(x) = a^2 e^{-ax}$

$$\Rightarrow F_c[f''(x)] = a^2 F_c[e^{-ax}]$$

$$FT- (19)$$

$$\Rightarrow -\omega^2 F_c(e^{-ax}) - \sqrt{\frac{2}{\pi}} f'(0) = a^2 F_c(e^{-ax})$$

$$\Rightarrow (a^2 + \omega^2) F_c(e^{-ax}) = a \sqrt{\frac{2}{\pi}} \quad [\because f'(0) = -a]$$

$$\Rightarrow F_c(e^{-ax}) = \left(\frac{a}{a^2 + \omega^2} \right) \sqrt{\frac{2}{\pi}}$$

Convolution theorem

Convolution: Convolution of two functions $f(x)$ and $g(x)$ denoted by $f * g$ is defined as $f * g = \int_{-\infty}^{\infty} f(u)g(x-u)du$

Convolution theorem

$$F(f * g) = \sqrt{2\pi} F[f(x)] F[g(x)]$$

1. Verify convolution theorem for $f(x) = e^{-x}$, $x > 0$ and $g(x) = x$, $x > 0$

$$\begin{aligned} \text{Soln: } F[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-x} e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-(1+i\omega)x}}{-(1+i\omega)} \right]_0^{\infty} = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{1+i\omega} \right) \end{aligned}$$

$$\begin{aligned} F[g(x)] &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} x \cdot e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} \left[x \left(\frac{e^{-i\omega x}}{-i\omega} \right) + \frac{e^{-i\omega x}}{\omega^2} \right]_0^{\infty} \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{-1}{\omega^2} \right) \end{aligned}$$

$$\sqrt{2\pi} F[f(x)] F[g(x)] = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{1+i\omega} \right) \left(\frac{-1}{\omega^2} \right)$$

FT (20)

$$f * g = \int_0^x f(u) g(x-u) du = \int_0^x e^{-u} (x-u) du$$

$$= \left[(x-u) \left(\frac{e^{-u}}{-1} \right) + e^{-u} \right]_0^x$$

$$= \underline{\underline{e^{-x} + x - 1}}$$

$$F(f * g) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} (e^{-x} + x - 1) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \left(e^{-(1+i\omega)x} + x e^{-i\omega x} - e^{-i\omega x} \right) dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-(1+i\omega)x}}{-(1+i\omega)} + x \frac{e^{-i\omega x}}{(-i\omega)} + \frac{e^{-i\omega x}}{\omega^2} + \frac{e^{-i\omega x}}{i\omega} \right]_0^{\infty}$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{1+i\omega} - \frac{1}{\omega^2} - \frac{1}{i\omega} \right] = \frac{1}{\sqrt{2\pi}} \left[\frac{-1}{(1+i\omega)\omega^2} \right]$$

Therefore $\underline{\underline{F(f * g) = \sqrt{2\pi} F[f(x)] F[g(x)]}}$

To find inverse fourier transform using convolution theorem

$$F(f * g) = \sqrt{2\pi} F[f(x)] F[g(x)]$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}} f * g = \bar{F}^{-1} [F[f(x)] F[g(x)]] = \bar{F}^{-1} [\hat{f}(\omega) \hat{g}(\omega)]$$

1. Find $\bar{F}^{-1} \left[\frac{1}{6 + 5i\omega - \omega^2} \right]$

Soln :- $\bar{F}^{-1} \left[\frac{1}{6 + 5i\omega - \omega^2} \right] = \bar{F}^{-1} \left[\frac{1}{(\omega + 2)(\omega - 3)} \right]$

$$= \bar{F}^{-1} [\hat{f}(\omega) \hat{g}(\omega)] \text{ where}$$

FT- (Q1)

$$\hat{f}(\omega) = \frac{1}{2+i\omega}$$

$$\hat{g}(\omega) = \frac{1}{3+i\omega}$$

$$\Rightarrow g(x) = F^{-1}(\hat{g}(\omega))$$

$$\Rightarrow f(x) = F^{-1}(\hat{f}(\omega)) = F^{-1}\left(\frac{1}{2+i\omega}\right)$$

$$= F^{-1}\left(\frac{1}{2+i\omega}\right)$$

$$= \sqrt{2\pi} e^{-2x}, \quad x > 0$$

$$= \sqrt{2\pi} e^{-3x}, \quad x > 0$$

$$\left[\text{Since } F(e^{-ax}, x > 0) = \frac{1}{\sqrt{2\pi}} \frac{1}{a+i\omega} \right]$$

$$\Rightarrow F^{-1}\left(\frac{1}{a+i\omega}\right) = \sqrt{2\pi} e^{-ax}, \quad x > 0$$

$$\therefore F^{-1}(\hat{f}(\omega)\hat{g}(\omega)) = F^{-1}\left(\frac{1}{(2+i\omega)(3+i\omega)}\right)$$

$$= \frac{1}{\sqrt{2\pi}} f * g$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^x \sqrt{2\pi} e^{-2u} \cdot \sqrt{2\pi} e^{-3(x-u)} du$$

$$= \frac{2\pi}{\sqrt{2\pi}} e^{-3x} \int_0^x e^u du$$

$$= \sqrt{2\pi} e^{-3x} (e^x - 1)$$